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Review

The value of a broad temporal and spatial perspective in understanding dynamics of kelp forest ecosystems

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Abstract. Many ecological processes play out over longer time scales and larger spatial scales than can be studied in a traditional 2–4-year grant cycle. Uncertainties in future funding hinder efforts to implement comprehensive research programs that integrate coupled time series observations of physical variables and ecological responses, manipulative experiments and synthetic analyses over the long term. Such research is essential for advancing our understanding of ecological responses associated with climate change, and the physical and biological processes that control them. This need is perhaps greatest for ecosystems that display highly dynamic and spatially complex patterns that are difficult to explain with short-term, small-scale studies. Such is the case for kelp forest ecosystems, which often show tremendous spatial and temporal variability in resource supply, consumer control and physical disturbance across spatial scales of metres to hundreds of kilometres and temporal scales of hours to decades. Here we present four examples from the Santa Barbara Coastal Long-term Ecological Research project that demonstrate the value of a broad temporal and spatial perspective in understanding the causes and ecological consequences of short-term local dynamics of giant kelp forests of California, USA.

Additional keywords: climate change, community structure and dynamics, human impacts, long-term ecological research, marine protected areas, net primary production, stable isotopes, trophic relationships.

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Introduction

There is broad scientific consensus that the Earth's climate system is changing at an unprecedented rate in response to increased anthropogenic activities (Stocker *et al.* 2013). The atmosphere and ocean have warmed, the amounts of snow and ice have diminished, sea level has risen, and the concentrations of greenhouse gases have increased. The unequivocal evidence for these changes is based on direct measurements dating back to the mid-19th century, remote sensing since the mid-20th century, and palaeoclimate reconstructions that extend back hundreds to millions of years. Together, these data provide a comprehensive view of the variability and long-term changes in the physical attributes of the atmosphere, the ocean, the cryosphere, and the land surface.

By comparison, our knowledge of long-term changes in the biological attributes of the Earth's systems is rather poor. Direct measurements of long-term changes in distribution and abundance are available for few species at a limited number of locations (Strayer *et al.* 1986; Lindenmayer *et al.* 2012), as are indirect measurements obtained from remote sensing (Kerr and Ostrovsky 2003; Cavanaugh *et al.* 2014) and palaeoclimate reconstruction (Kennett and Stott 1991; Behl and Kennett 1996). Long-term records useful for evaluating climate and anthropogenic-related changes in trophic structure, species interactions and ecosystem processes such as energy flow, nutrient cycling, and material export are even more scarce (but see Brock *et al.* 2007). Moreover, most predictions concerning the ecological consequences of future changes in ocean temperature and chemistry are based in large part on controlled laboratory experiments that are unable to fully replicate the natural variability of the physical environment and the complexity of species interactions that occur in nature. Suffice it to say that our understanding of the ecological consequences associated with the Earth's changing climate and expanding human influence is much more limited than our understanding of the physical and sociological basis for climate change itself.

Ecology is a long-term, broad-scale science, as many ecological patterns and processes occur over time scales of decades to centuries and spatial scales of tens to thousands of kilometres. However, there are few opportunities for ecologists to conduct long-term broad-scale research as most ecological studies are funded for short periods with a limited spatial and programmatic scope. Consequently, most long-term efforts in the ecological sciences have not been planned. Instead they tend to be

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somewhat narrow in scope and pieced together in a sequential fashion by dedicated individuals with the vision and entrepreneurial skills needed to sustain a long-term program that typically ends with their career (Strayer *et al.* 1986; Lindenmayer *et al.* 2012). This type of research model, although valuable, is often inadequate for detecting and understanding the ecological consequences of alterations in the environment due to climate change and a plethora of increasing human activities (Strayer *et al.* 1986). For this purpose planned long-term ecological research is needed that couples coordinated time-series data of ecological responses and their abiotic drivers with experiments and synthetic analyses specifically aimed at isolating the causes of ecological change.

The US National Science Foundation recognised the need for a more stable platform for the ecological sciences in 1980 when it established its Long-term Ecological Research (LTER) Network, which presently consists of 25 sites representing a diverse array of ecosystems. The intent of this program was not to initiate long-term ecological monitoring, but rather to develop long-term site-based research programs with the objective of attaining a predictive understanding of ecological phenomena over long temporal and large spatial scales (Callahan 1984). This concept has grown globally and in 1993 the International Long-term Ecological Research (ILTER) Network, a 'network of networks', was founded with the explicit purpose of promoting an increased understanding of environmental change across the globe. As of 2013, 40 member national or regional networks comprising hundreds of sites around the globe have established formal LTER programs and joined the ILTER Network.

Systems where the need for long-term ecological research is perhaps greatest are those that not only provide valuable goods and services to society, but also fluctuate greatly in response to a variety of physical and biological forces. Such system attributes makes distinguishing inherent natural variability from long-term secular trends associated with climate change and human actions more difficult. Shallow coastal reefs in temperate regions of the world are a system that exemplifies these attributes. Coastal zones in temperate latitudes are among the most populated areas of the world and shallow reefs in these regions are dynamic systems that provide myriad goods and services that fluctuate greatly in response to natural and human drivers (Beaumont *et al.* 2007; Rönnbäck *et al.* 2007; Barbier *et al.* 2011).

Temperate reefs dominated by macroalgae, especially large kelps, are among the most productive ecosystems on Earth (Mann 2000; Reed and Brzezinski 2009). The high productivity and three-dimensional structure provided by kelps supports diverse assemblages of animals, many of which have high economic value as food, pharmaceuticals, and other products (Foster and Schiel 1985; Andrew 1999; Radjasa *et al.* 2011; Carr and Reed, in press). Unlike long-lived slow-growing corals that dominate tropical reefs, most kelps live for only a few years and are among the fastest-growing autotrophs in the world (Schiel and Foster 2006; Graham *et al.* 2007; Reed and Brzezinski 2009). They often show low resistance to many forms of physical and biological disturbance, but high resilience with rapid recovery once disturbance has abated (Dayton 1985; Hawkins and Harkin 1985; Kennelly 1987; Schiel and Foster 1986; Tegner and Dayton 1987). As a result the ecological communities that inhabit temperate reefs tend to be very dynamic and fluctuate greatly in response to a wide variety of natural and anthropogenic perturbations. Thus a long-term, broad-scale perspective that captures short-term, small-scale dynamics is essential for detecting secular trends in kelp forest communities associated with environmental change. Indeed, some of the most important insights into the response of temperate reef communities to environmental forcing have come from broad spatial and temporal data assembled by a small number of research and monitoring programs around the globe (Helmuth *et al.* 2006; Hawkins *et al.* 2008, 2009; Parnell *et al.* 2010, Smale *et al.* 2010, Kenner *et al.* 2013; Kushner *et al.* 2013).

Sustained funding has allowed the Santa Barbara Coastal LTER project to develop an integrated research program aimed at developing a predictive understanding of local and regional controls of community structure and ecosystem dynamics of giant kelp (*Macrocystis pyrifera*) forests. Below we present four examples that demonstrate the value of a broad spatial and temporal perspective in understanding the ecology of giant kelp forest communities on shallow reefs off the coast of California, USA.

Patterns and controls of net primary production

Much of what we know about the productivity of temperate reef systems has come from short-term studies conducted over small spatial scales using a wide variety of methods that frequently measure different attributes (Littler and Arnold 1980). Consequently, estimates of macroalgal standing biomass and net primary productivity (NPP) for temperate reefs are as numerous and variable as the methods that have been devised to measure them (Alongi 1998). Time series data of macroalgal standing biomass and NPP needed to assess the consequences of long-term change require frequent sampling to capture short-term changes in these dynamic systems, which are best obtained using non-destructive methods that allow the same plots to be repeatedly sampled over time. The logistical difficulties associated with such high-frequency sampling across entire reefs has led to few datasets of macroalgal biomass and NPP from which the consequences of long-term change can be assessed.

Non-destructive methods of estimating standing biomass and NPP of giant kelp and understorey algae in giant kelp forests off California, USA, have been developed by the Santa Barbara Coastal LTER project (Rassweiler *et al.* 2008; Reed *et al.* 2009; Miller *et al.* 2012), and time series data using these methods have provided novel insights into the patterns and drivers of variability in macroalgal biomass and NPP at both local and regional scales (Reed *et al.* 2008, 2011; Harrer *et al.* 2013). NPP generally signifies the dry mass of plant matter (or plant carbon) produced per unit area of the Earth's surface per unit time (Reichle *et al.* 1975; Webb *et al.* 1983; Fahey and Knapp 2007), and it is a function of the density of actively growing plant mass and its rate of increase due to growth and recruitment. Counter to conventional wisdom based on short-term observations, we found that growth did not explain any of the year-to-year variation of NPP by giant kelp in our long-term plots off Santa Barbara (Reed *et al.* 2008). Instead, 84% of the observed interannual variation in NPP was explained by the standing

Table 1. Time series data collected from southern and central California over a 9-year period

Data are means averaged over all sites ($n = 9$ in southern California, $n = 8$ in central California). From Reed *et al.* (2011)

	Central California	Southern California
Nutrients (NO_3 μmol L^{-1})	6.65	1.70
Grazing (number of urchins m^{-2})	0.07	7.07
Significant wave height (m)	2.2	1.2
Annual canopy loss from waves (%)	93	51
NPP (g dry mass m^{-2} $year^{-1}$)	1.11	2.24

biomass of giant kelp at the beginning of the growth year ($R^2 = 0.63$) and the recruitment of new plants during the year ($R^2 = 0.21$). This result runs counter to most studies that investigate patterns and controls of variability in plant biomass and NPP, which have focussed on the availability of resources that stimulate growth and recruitment (i.e. bottom-up processes: Rosenzweig 1968; Webb *et al.* 1983) or interactions between predators and primary consumers that alter plant biomass (i.e. top-down processes: Hairston *et al.* 1960; Terborgh and Estes 2010).

Our results led us to hypothesise that storm-wave disturbances that remove giant kelp before the start of the growing season overwhelm bottom-up and top-down processes to act as the principal driver of NPP in giant kelp forests off California. To test this hypothesis we made use of known regional differences in environmental forcing and predator abundance in California kelp forests to examine the relative importance of grazing pressure (top-down), nutrient availability (bottom-up), and storm waves (disturbance) on the standing biomass and NPP of giant kelp (Reed *et al.* 2011). We used time series data collected from nine sites in southern California and eight sites in central California over a 9-year period to show that there are consistent differences in nutrients, grazers and waves between central and southern California kelp forests (Table 1). These differences indicate that top-down control should cause reefs in central California to support more kelp biomass and higher production than reefs in southern California because of the near absence of sea urchin grazers in central California due to the presence of sea otters, a keystone predator. Bottom-up control should also lead to more kelp in central California due to kelp's rapid growth in the cold, nutrient-rich waters of central California, and slower growth on the more oligotrophic reefs of southern California. Surprisingly, we found that patterns of kelp biomass and production did not match either of these predictions. Instead we found that giant kelp NPP was consistently lower in central California, most likely because more intense wave disturbance on the central coast overwhelmed top-down and bottom-up forces (Table 1).

Results derived from satellite remote sensing used to extend the spatial and temporal scales of our diver estimates are consistent with this conclusion. We developed a novel method for estimating the canopy biomass of giant kelp from Landsat 5 Thematic Mapper satellite imagery, which allowed us to examine the dynamics of giant kelp biomass on spatial scales ranging from hundreds of square metres to hundreds of square kilometres and temporal scales ranging from several weeks to >25 years (Cavanaugh *et al.* 2011). This method relied on calibrating the reflectance of giant kelp in Landsat images to long-term *in situ* measurements of kelp canopy biomass at three sites performed by divers each month since 2002 (Rassweiler *et al.* 2008). Using this method we found high spatial variability in kelp biomass dynamics within the Santa Barbara Channel that was linked to spatial variability in environmental conditions (e.g. wave height, sea surface temperature) (Cavanaugh *et al.* 2011). Subsequent analyses showed that synchrony in the biomass dynamics of giant kelp at small spatial scales (50 m to 1.3 km) corresponded to the abundance of sea urchins and young kelp recruits, suggesting that local drivers of consumer pressure (top-down) and resource availability for recruitment (bottom-up) influence small-scale synchrony in kelp populations (Cavanaugh *et al.* 2013). Kelp synchrony at larger scales (1.3–172 km) corresponded to physical drivers (especially wave height), suggesting that regional environmental variability in wave disturbance dominates population dynamics at larger spatial scales.

Drivers of community structure and dynamics

Facilitation and competition are known to play important roles in structuring ecological communities (Underwood 1978; Connell 1983; Bertness and Shumway 1993). Indirect positive effects of one species on another by suppression of a common competitor (termed indirect facilitation) can be stronger than the direct effects of competition (Miller 1994; Levine 1999; Callaway and Pennings 2000; Pagès *et al.* 2003). Not surprisingly, spatial variation in the abundance of species that are indirect facilitators is known to influence the distribution and abundance of associated species engaged in competition (Miller 1994; Levine 1999; Callaway and Pennings 2000). Much less is known about the dynamical consequences of temporal fluctuations in the abundance of an indirect facilitator. This phenomenon is best examined using time series data of abundances of competing species and their indirect facilitator.

Interactions involving indirect facilitation are thought to be most widespread in assemblages where pairs of competing species have few resources in common (Levine 1999). Such is the case for shallow temperate reefs where the primary space holders (i.e. macroalgae and sessile suspension-feeding invertebrates) occupy different trophic levels. Macroalgae are primary producers that derive their nutrition from sunlight and dissolved nutrients, whereas sessile invertebrates are consumers that are nourished by filtering plankton and other organic matter from the water column. Variation in unshared resources such as light or plankton can determine the outcome of space competition between macroalgae and sessile invertebrates.

In giant kelp forests, light available to understorey algae is greatly influenced by shading from the canopy of the giant kelp, which fluctuates greatly in space and time due to physical disturbance, grazing and variable growth conditions (Foster and Schiel 1985; Graham *et al.* 2008; Carr and Reed, in press). These fluctuations can have profound effects on the structure of the benthic community. Using field experiments we demonstrated a negative effect of giant kelp on light availability and

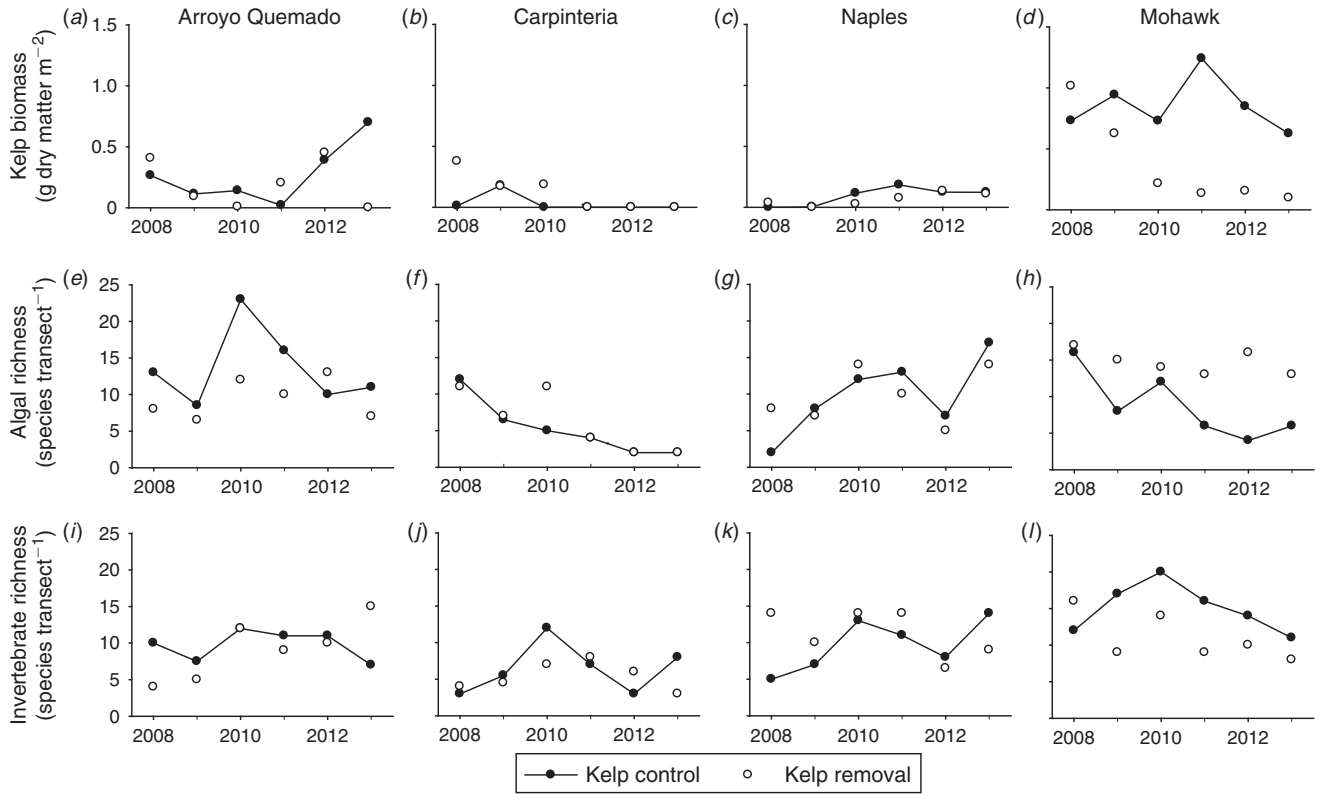

Fig. 1. Summertime values for the biomass of giant kelp, species richness of macroalgae, and species richness of macroinvertebrates in kelp-control and kelp-removal plots at four kelp forest sites (Arroyo Quemado, Carpinteria, Naples, Mohawk) for the period 2008–13. Giant kelp was removed once per year in the winter in the kelp-removal plots.

understorey algal abundance and a positive effect on the abundance of sessile invertebrates, which was consistent with an indirect effect mediated by shade from the kelp canopy (Arkema *et al.* 2009). Results from Structural Equation Modelling (SEM) using data collected at 16 locations over an 8-year period revealed that the magnitude of the indirect positive effect of giant kelp on sessile invertebrates was six times larger than its direct effect on them (Arkema *et al.* 2009). These results indicate that the dynamics of this system are driven by variability in the abundance of a single structure-forming species that has indirect positive, as well as direct negative, effects on associated species.

Subsequent analyses of these time series data using SEM showed that the structure of the kelp forest food web was linked to the frequency of winter storms that removed giant kelp (Byrnes *et al.* 2011). Periodic storms helped to maintain the diversity (measured as species richness) and complexity (measured as the mean number of feeding links per species) of the food web by removing giant kelp and stimulating the growth of understorey algae and the consumers that associate with them. Results from SEMs were borne out in modelled simulations, as food web diversity and complexity were greater in forests subjected to 1 year of large waves compared with those subjected to the minimum level of observed wave disturbance. Interestingly, modelled simulations also showed that a higher annual frequency of severe winter storms (as predicted by some climate models) leads to less diverse and less complex food webs with fewer trophic levels and fewer feeding links per species. Byrnes *et al.* (2011) tested the hypothesis that an increase in the frequency of severe storms will simplify kelp forest food webs using a multisite kelp-removal experiment designed to simulate frequent large storms. Initial results from the experiment were consistent with the predictions of the model simulations in that species richness and linkage density of the food web were higher at sites where more kelp was removed (either experimentally or naturally by storms) in the first year, but not the second year, after two successive years of kelp removal.

We have continued this multisite experiment by removing giant kelp from one of two paired 2000-m^2 plots at four reefs each winter since 2008, and have followed changes in community composition within each plot over the course of the intervening 6 years. Results obtained to date (2008–13) were not anticipated, and underscore the value and need for multisite, multiyear experiments. At three of our four study reefs (Arroyo Quemado, Carpinteria and Naples) we have yet to observe a significant difference in the summertime biomass of giant kelp between our winter kelp-removal and kelp-control plots because a combination of other factors (e.g. sedimentation, grazing by sea urchins, wave disturbance) have acted indiscriminately to reduce giant kelp in both kelp-removal and kelp-control plots (Fig. 1*a–c*). As a result, we have yet to see significant differences in the number of species of algae and invertebrates between the kelp-removal and kelp-control plots at these three

reefs (Fig. 1*e–g*, *i–k*). In contrast, we have witnessed a steady decline in the summertime biomass of giant kelp in the winter kelp-removal plot relative to the kelp-control plot at the fourth reef, Mohawk (Fig. 1*d*). Understorey algae have increased in abundance over time in the kelp-removal plot at Mohawk and have prevented giant kelp from reinvading. Subsequent changes in the species richness of algae and invertebrates observed at Mohawk are consistent with the findings of Arkema *et al.* (2009) and Byrnes *et al.* (2011) in that algal species richness is higher and invertebrate species richness is lower in the kelp-removal plot compared with the kelp-control plot (Fig. 1*h*, *l*). These results highlight the variable nature of giant kelp ecosystems and the value of a broad spatial and temporal perspective when evaluating processes that influence their dynamics.

Trophic relationships and energy flow

Food webs are useful descriptions of trophic relationships, but in general most webs have not incorporated variation in time or space. Fluctuating productivity, migrations, and pulsed subsidies are some of the processes that can cause profound changes in sources of production supporting food webs over both time and space (Sears *et al.* 2004). Stable isotope analysis is a potentially powerful approach for evaluating the trophic pathways that comprise food webs. Basal sources (i.e. primary producers) can be identified if they have distinct isotopic signatures that change predictably as they move through the food web. Carbon isotope values ($\delta^{13}C$, ‰) of consumers reflect the value of dietary carbon (DeNiro and Epstein 1978; Fry and Sherr 1984), and nitrogen isotope values can also help to identify basal sources as well as trophic position (Fry 1988; Post 2002; Page *et al.* 2013). However, the interpretation of carbon and nitrogen isotope data is often hindered by a lack of information on temporal and spatial variability in the isotope values of producers and consumers. Time series data on the abundance and isotope values of basal sources from multiple locations are invaluable in identifying basal source contributions to food webs and their variability in time and space.

Sessile invertebrates that depend on suspended particulate organic matter (POM) for food are an abundant group of primary consumers on shallow coastal reefs worldwide (Gili and Coma 1998; Witman and Dayton 2001). In highly productive temperate regions, the source of organic matter sustaining these benthic consumers is poorly known. Marine ecologists have taken advantage of stable isotope analysis by using mixing models to estimate contributions of kelp detritus to coastal temperate food webs across the globe (e.g. Duggins *et al.* 1989; Fredriksen 2003; Kaehler *et al.* 2006). Many of these studies have asserted that the contribution of kelp-derived POM to the diet of suspension-feeders approaches or exceeds that of phytoplankton (reviewed by Miller and Page 2012). Such assertions have relied on short-term efforts to sample consumers and basal sources, typically kelp and phytoplankton. A $\delta^{13}C$ value for phytoplankton is typically derived from offshore samples or the literature and is well separated from kelp values, enabling the use of mixing models to calculate the relative contributions of these two different basal sources. An important benefit of isotopes is that they integrate the incorporation of food sources over time, making data from short-term sampling efforts representative of longer periods for consumers (Schindler and Lubetkin 2004). However, isotope values of primary producers and detrital energy sources can vary substantially with time (Post 2002). Such high temporal variability complicates the use of isotopes in identifying the basal sources used by secondary consumers, which integrate isotope values over much longer periods (Cabana and Rasmussen 1996; Post 2002). This is particularly true for phytoplankton, whose populations are highly dynamic (Anderson *et al.* 2008) and variable in carbon isotope composition (Fry and Wainwright 1991; Laws *et al.* 1995).

Table 2. Mean and range of stable carbon and nitrogen isotope values for the major sources of primary production in giant kelp forests and for reef suspension feeders
From Page *et al.* (2008)

	$\delta^{13}C$ (‰)	$\delta^{15}N$ (‰)
Giant kelp	−12.9 (−13.8 to −12.2)	9.3 (8.5 to 9.7)
Reef POM	−19.1 (−23.7 to −14.1)	8.6 (2.7 to 14.5)
Offshore POM	−21.0 (−22.4 to −19.6)	6.8 (5.7 to 7.5)
Terrestrial POM	−24.6 (−26.3 to −22.9)	2.9 (−1.6 to 4.5)
Reef suspension feeders	−16.9 (−18.7 to −15.5)	12.1 (11.1 to 13.2)

Time series data on basal sources and consumers collected by Santa Barbara Coastal LTER researchers have stimulated new interpretations of patterns of isotope values in reef consumers. Page *et al.* (2008) sampled carbon and nitrogen stable isotopes monthly for 4–5 years at four reefs off Santa Barbara to examine the relationships between producer abundance and isotopic signatures of producers and consumers, including suspension feeders. The abundances and isotopic signatures of primary producers were found to be quite variable over time and space (Table 1). The suspended POM comprising the food source for suspension feeders on the reefs is potentially a mixture of different carbon sources with different isotope signatures, including living and detrital phytoplankton as well as detritus from giant kelp and other reef macroalgae and land plants. Reef POM had a $\delta^{13}C$ signature that was intermediate between that of giant kelp and offshore POM, a pattern that almost invariably has been interpreted as the result of the mixing of phytoplankton and kelp detritus. $\delta^{13}C$ of reef suspension feeder tissues, unsurprisingly, matched up well with those of reef POM (Table 2). The same pattern has been seen in many isotope studies of coastal marine ecosystems across the world (reviewed in Miller and Page 2012). The interpretation of this pattern has also been invariant: particulate kelp detritus contributes significantly to reef POM, in turn providing a significant source of dietary carbon to sessile invertebrates in the kelp forest. What most of these studies have lacked, however, is a temporal perspective. Using time series data, Page *et al.* (2008) were able to test this general hypothesis in a different way. If kelp contributes significantly to the dietary carbon of reef suspension feeders, then we predicted that the $\delta^{13}C$ of reef POM would be correlated with kelp biomass over space and time. This prediction was not supported as no relationship was found between the amount of kelp biomass on a reef and $\delta^{13}C$ of reef POM (Fig. 2). Instead, $\delta^{13}C$ of reef POM was generally correlated with the concentration of chlorophyll *a* in the water column, suggesting that enriched POM $\delta^{13}C$ values were associated with

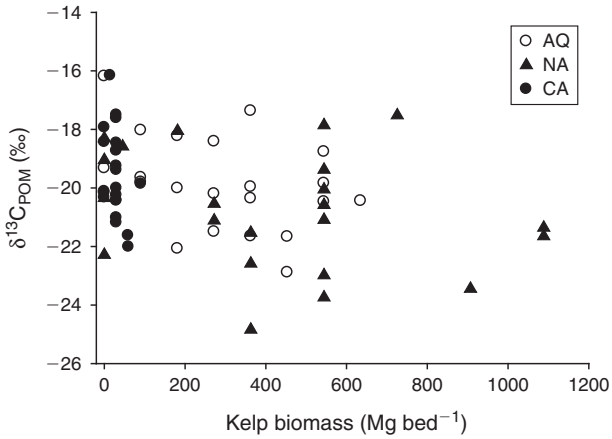

Fig. 2. Relationship between the standing biomass of giant kelp and the $\delta^{13}C$ value of suspended POM at three reef sites (AQ, Arroyo Quemado; NA, Naples; CA, Carpinteria) from 2002 to 2005. From Page *et al.* (2008).

phytoplankton blooms rather than inputs of kelp detritus. Isotopic signatures of benthic herbivores were more enriched than those of suspension feeders, reflecting a diet of kelp and other benthic macroalgae. This result suggested that suspended kelp particles do not contribute significantly to reef POM, and hence are not a significant food source for kelp forest suspension feeders.

Subsequent studies have supported this interpretation; variability in the $\delta^{13}C$ of reef POM was well explained by phytoplankton production, and enriched values of $\delta^{13}C$ occurred during phytoplankton blooms (Miller *et al.* 2013). Production of particulate organic carbon (POC) ≤250 µm (the size fraction available to suspension feeders) by kelp was measured and found to contribute <0.2% of the total suspended reef POC in the kelp forest (Yorke *et al.* 2013). Collectively these findings suggest that, counter to common dogma (e.g. Steneck *et al.* 2002; Graham 2004; Estes *et al.* 2011; Smale *et al.* 2013), phytoplankton, and not suspended kelp detritus, are the primary food source for kelp forest suspension feeders. Time series sampling across sites varying in kelp production enabled this transformative interpretation. It should be noted that an increase in the spatial extent of sampling could perhaps substitute for time with regard to characterising variability in isotope values of suspension feeders relative to kelp abundance, because kelp standing biomass varies substantially among reefs (Dayton *et al.* 1984; Edwards 2004; Reed *et al.* 2006). Nevertheless, space would likely be a poor substitute for time in sampling and characterising variability in phytoplankton, because phytoplankton blooms vary in concert with seasonal upwelling and other oceanographic patterns (e.g. Brzezinski and Washburn 2011). These considerations suggest that our ability to elucidate environmental change in food webs will benefit greatly from broad spatial and temporal sampling to account for the wide range in population turnover times and patch dynamics among species in an ecosystem.

Assessing management impacts in dynamic systems

A major goal of ecological research is to understand how human activities affect ecosystems, and in particular to measure the outcome of management actions. For example, in southern California kelp forests managers have recently established a network of marine protected areas (MPAs) in which areas of the coastline are closed to fishing (Airamé *et al.* 2003; Saarman *et al.* 2013). Measuring the outcome of MPA establishment is important both to determine whether the MPAs are working as intended, and because such a dramatic change in human activities provides a valuable opportunity for understanding the dynamics of the system. Although there has been considerable theoretical work predicting the outcome of MPAs in southern California (Rassweiler *et al.* 2012; White *et al.* 2013), evaluating their effectiveness empirically requires patience, as some effects may take many years to occur (Babcock *et al.* 2010).

Although many of the MPAs in southern California are too new to evaluate empirically, a group of MPAs at the islands on the south side of the Santa Barbara Channel are now more than a decade old (they were established in 2003). The direct effect of these MPAs was to reduce harvests of targeted fish and invertebrates, and there is accumulating evidence that they have increased populations of those species (Hamilton *et al.* 2010; Kay *et al.* 2012). Protected areas within the region have also been proposed to increase non-targeted species through indirect effects. Most notably, increases in the abundance of fished predators within protected areas have been proposed to reduce herbivores and increase the prevalence of kelp-dominated community states (Behrens and Lafferty 2004; Shears *et al.* 2012).

Unfortunately, data to evaluate marine protected areas are rarely collected before implementation. For example, three protected areas were established at Santa Rosa Island, which has historically supported a little more than half of the giant kelp (*Macrocystis pyrifera*) biomass at the four northern Channel Islands (over the past 50 years, kelp beds at Santa Rosa Island have averaged 56% of the total biomass of giant kelp at the four islands combined: Reed 2010). Although the United States National Park Service had three long-term monitoring sites at Santa Rosa Island at the time the MPAs were established, none were located in the areas chosen for protection, meaning there are no baseline monitoring data for those MPAs. To measure the effects of the MPAs, three new monitoring sites surveyed annually by divers were established by the National Park Service in 2005, 2 years after protection was implemented (see Kushner *et al.* 2013). The increases in giant kelp abundance at these sites suggest an indirect positive effect of reserve protection on giant kelp (Fig. 3*a*), but our ability to infer this is impaired by lack of data from before MPA establishment.

Fortunately, other longer-term data on giant kelp are available from the region, including satellite-based remote sensing (described in 'Patterns and controls of net primary production' above) and aerial surveys by the kelp-harvesting industry (Reed 2010). The satellite estimates support the divers' observations of increasing kelp after MPA establishment, but also show that kelp was increasing at the island even before reserve establishment (Fig. 3*b*). Further, these data emphasise the high degree of variability of the system. This variability is even more dramatically apparent in the aerial survey data collected by kelp harvesters. Over the five decades surveyed, major increases in giant kelp biomass happened multiple times.

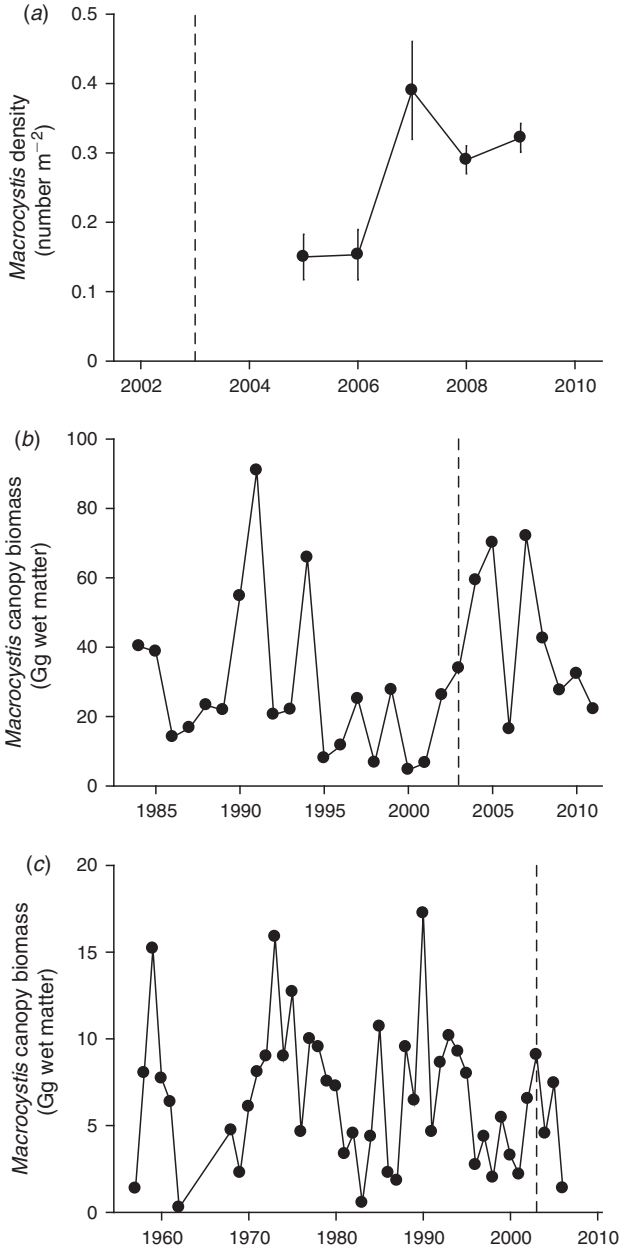

Fig. 3. Abundance of giant kelp (*Macrocystis pyrifera*) at Santa Rosa Island. (*a*) Mean (±s.e.) plant density in MPAs at Santa Rosa Island as measured by divers during summer. (*b*) Yearly maximum canopy biomass at Santa Rosa Island estimated from Landsat 5 thematic mapper imagery. (*c*) Yearly maximum canopy biomass at Santa Rosa Island estimated from aerial surveys by kelp harvesters. In each panel, the vertical dashed line indicates the year of MPA establishment at Santa Rosa Island.

Indeed, the magnitude of the recent increases in kelp is small relative to that which occurred before the establishment of the MPAs (Fig. 3*c*).

These longer-term datasets emphasise the importance of understanding reef community dynamics within a broad spatial and temporal scale. Kelp forest systems are naturally prone to sudden changes in species abundance either because of non-linear relationships between long-period climatic drivers and biological processes (Hsieh *et al.* 2005), or because positive feedbacks between key species can lead to stable periods punctuated by sudden dramatic change (Harrold and Reed 1985; Rassweiler *et al.* 2010; Ling *et al.* 2009). This is particularly true for kelp forests off southern California where the dominant kelp, *Macrocystis pyrifera*, is short lived (Graham *et al.* 2007), and the major environmental drivers of community dynamics (e.g. upwelling, El Niño, wave disturbance, sea urchin grazing) are irregular in their occurrence and intensity (North and Pearse 1970; Tegner and Dayton 1987; Dayton *et al.* 1999; Reed *et al.* 2011). When evaluating the effects of management actions or other human impacts, a long-term perspective is crucial to determining what represents a significant event driven by human activities and what might just as easily be inherent natural variability of the system.

Conclusions

Because of their dynamic nature, temperate reefs typically require high-frequency, long-term study to fully understand their response to changes in climate and human actions. Despite the global significance of these iconic systems few research programs have the capability to rigorously evaluate the extent to which they are responding to directional environmental change, especially in subtidal systems dominated by kelps. This is not due to a lack of desire by temperate reef ecologists, but instead a dearth of funding for long-term research. Research programs such as LTER are designed specifically for this purpose. The intent of LTER is to develop comprehensive peer-reviewed research programs with long-term objectives aimed at developing a predictive understanding of ecological phenomena over broad spatial scales. This is made possible by stable funding platforms that ensure research continuity and data preservation. A long-term perspective necessitates research that extends beyond the lifespans of its investigators, which requires rigorous documentation of methods and data and careful planning for leadership transition. The goal should not be to make data the exclusive property of the scientists who collect them, but rather to have the data regarded as a valuable resource by the scientific community at large (Callahan 1984). This requires a strong emphasis on data integrity and an altruistic philosophy aimed at maximising data sharing and accessibility. Electronic archives that assign digital objective identifiers (DOI) to individual datasets not only improve access to data, but provide a mechanism for acknowledging the individuals who collected them.

There are now a wide range of ecosystem types represented by the hundreds of sites in the ILTER Network. Unfortunately, only a handful of these sites focus on temperate reefs, and to our knowledge the Santa Barbara Coastal LTER is the only LTER site that focuses on kelp forests. The ecological and socio-economic value of kelp forests is widely recognised (Foster and Schiel 1985; Andrew 1999; Steneck *et al.* 2002; Carr and Reed, in press) and there is a clear and compelling need for comprehensive long-term broad-scale studies to understand their complex ecology. Increasing the number of long-term ecological research programs that target temperate reefs generally, and kelp forests in particular, is sure to pay great dividends towards

improving our understanding of the causes and consequences of ecological change now and into the future.

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